

Diego Martinez

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EDUCATION

The University of Texas at El Paso (UTEP)

Masters of Science in Artificial Intelligence

Anticipated: 12/2027

Overall GPA:4.0/4.0

Relevant coursework: Deep Learning, Machine Learning, Advanced Algorithms, Decision Making, Probability and Statistics

Campus Involvement: Google Developer Student Club, Society of Hispanic Professional Engineers, The Coding Interview Club

The University of Texas at El Paso (UTEP)

Bachelor of Science in Computer Science concentration in Data Science, Minor in Math

09/2023 - 05/2026

Overall GPA:3.91/4.0

Honors and Awards: College of Engineering Dean's List, UTEP Presidential Scholarship, Magna Cum Laude

SKILLS

- **Programming Languages:** Python, Java, JavaScript, C, C #, C++, MySQL, SQL, Dart, PHP, Bash, Zsh
- **Frameworks:** React.js, Node.js, Angular.ts, Flask
- **Tools:** Firebase, AWS Console, Azure Portal, Git, VS Code, GitHub, Docker, Supabase, HTML, CSS, UNIX, Figma, LLM's
- **Libraries:** Pandas, Scikit-learn, NumPy, Matplotlib, PyTorch, FAISS, spaCy, NLTK, OpenCV

EXPERIENCE

UNIVERSITY OF TEXAS AT EL PASO

12/2025 - Current

ALFA:UNDERGRADUATE RESEARCH ASSISTANT

- Conducted systematic experiments evaluating **ALFA-M and ALFA-ALL** across **seven active learning functions** (Random, PowerMargin, Entropy, CLUE, Margin, Density, GALAXY) on **CIC-IDS cybersecurity datasets**, achieving **84% F1-score**.
- Executed **multi-seed experimental runs** (42, 123, 456) to validate model robustness and statistical significance, computing **mean ± standard deviation** metrics and identifying **ALFA-M's superior performance** over ALFA-ALL in low-label regimes.
- Benchmarked **generative augmentation models** (CTGAN, TVAE, RTF) and **baseline methods** to evaluate their impact on **class-imbalanced tabular data**, contributing to reproducible active learning research under **distribution shift**.

UNIVERSITY WRITING CENTER/MINER LEARNING CENTER

01/2025 - Current

FRONT-END DEVELOPER

- Improved user navigation and access to information by **120%** through developing a search feature using **React** and **JavaScript**, allowing users to find supported courses, professors, assigned tutors, tutor office hours, and tutoring schedules.
- Increased the visit and retention of the website **40%** by implementing engaging and responsible user interface features with **JavaScript** and **React**.

LEADERSHIP

GOOGLE DEVELOPER GROUP AT UTEP

08/2025 - Current

TECHNICAL OFFICER / MENTORSHIP PROGRAM COORDINATOR

- Organized and delivered technical workshops on **Angular**, **Android Studio**, **Firebase**, **Flutter**, and **Google Cloud** for 100+ attendees, creating step-by-step installation guides, **troubleshooting resources**, and curating a **repository** of tutorials.
- **Directed** a Mentorship Program for 90+ students, pairing mentors with mentees to provide **guidance** on **interview preparation**, **LinkedIn optimization**, leadership, and career planning, strengthening student **professional development**.

GOOGLE DEVELOPER GROUP MENTORSHIP PROGRAM

03/2025 - Current

MENTOR

- Guided two people with resume and LinkedIn optimizations, interview preparation, project development, and career advising supporting professional development and boosting confidence.

PROJECTS

RAG-POWERED STUDY ASSISTANT – PYTHON, CLAUDE API, FAISS, LANGCHAIN

02/2026

- Improved semantic search accuracy by 35% using a **RAG** backend backend using **LangChain**, **FAISS** vector database, and **Claude API** embeddings with advanced chunking, and hybrid retrieval for optimized document **recall**.
- Reduced hallucinations by 40% via structured **prompt engineering** templates with enforced JSON schema validation for consistent Q&A outputs.
- Built an autonomous **AI agent** layer for query decomposition and multi-hop reasoning, leveraging **NLP techniques** (named entity recognition, semantic similarity) to route complex questions across document sources.
- Accelerated model iteration by 50% by building an automated evaluation **pipeline** measuring relevance, faithfulness, and completeness for **data-driven refinement** of retrieval strategies.
- Reduced hallucinations by 40% via structured prompt engineering with enforced JSON schema validation for consistent Q&A outputs.

MACHINE LEARNING MUSIC GENRE CLASSIFIER – PYTHON, SCIKIT-LEARN, TENSORFLOW

11/2025

- Built and evaluated 5 **machine learning models** (Logistic Regression, KNN, Random Forest, SVM, DNN) to classify 106,574 audio tracks across 161 genres from the **FMA Large dataset**, achieving **67% accuracy** with a **Balanced DNN**.
- Preprocessed dataset by filtering to include only genres with 50+ samples per category to address **class imbalance**; implemented data pipeline using **Pandas** and **NumPy** for training/validation splits across the multi-class **classification task**.

BLOOMBERG TECH LAB ON CAMPUS – FLASK, REACT (Tech Stack)

02/2025

- Developed **full-stack** web application, leveraging **React** for the frontend, **Flask** for the backend, **Redis** for in-memory caching, and **Docker** for containerized deployment, ensuring a scalable and responsive user experience.